FOR EDITH · PAEDIATRICS & CHILD HEALTH JUNIOR CLERKSHIP

Infant & Young Child Feeding

Lecture | Week 5 · Measles, vaccines & the newborn | Paediatrics | Dr Namugga

Start breastfeeding within 1 hour of birth, give only breast milk for 6 months, then add safe, energy-dense and diverse complementary foods from 6 months while breastfeeding continues to 2 years or beyond; mothers living with HIV in Uganda breastfeed the same way on ART, and most "not enough milk" problems are fixed by correcting positioning and attachment.

Must know before the ward round

1. WHO/UNICEF/Uganda MoH: initiate breastfeeding within 1 hour of birth, exclusive breastfeeding (EBF) for 6 months, complementary foods from 6 months (180 days), continue breastfeeding to 2 years or beyond (WHO 2023 complementary feeding guideline).
2. Exclusive breastfeeding = breast milk only; ORS, drops and syrups (vitamins, minerals, medicines) are allowed. No water, glucose water, herbs or formula.
3. Good attachment (4 signs): more areola visible above than below the mouth, mouth wide open, lower lip turned outward, chin touching the breast. Good positioning (4 points): head and body in line, close to mother, facing the breast (nose to nipple), whole body supported.
4. Adequate milk intake: 8–12 feeds in 24 h, at least 6 wet nappies a day after day 5, birth weight regained by 10–14 days, steady weight gain on the growth chart.
5. Complementary feeding (UCG 2023/IMCI): 6–9 months 2–3 meals + 1–2 snacks, start with 2–3 tablespoons and increase to ½ cup (250 mL cup); 9–12 months ½ cup 3–4 meals + 1–2 snacks; 12–24 months ¾ cup 3–4 meals + 1–2 snacks; 2 years and over at least 1 full cup.
6. WHO 2023 complementary feeding: animal-source foods (meat, fish, eggs) daily, fruits and vegetables daily, pulses/nuts/seeds often; no sugary drinks, no foods high in sugar, salt or trans fats, no non-sugar sweeteners; limit 100% juice; responsive feeding.
7. Minimum dietary diversity = at least 5 of 8 food groups; minimum meal frequency = 2 (breastfed 6–8 months), 3 (breastfed 9–23 months), 4 (non-breastfed, including 2 milk feeds). Uganda: only about 15% of 6–23-month-olds get a minimum acceptable diet.
8. HIV in Uganda: mothers on ART exclusively breastfeed for 6 months, then complementary foods with continued breastfeeding; MIYCAN 2021 advises stopping at 12 months for HIV-negative exposed infants (with ≥ 500 mL/day of other milk) and continuing to 24 months if the infant is HIV-positive; WHO 2016: at least 12 months and up to 24 months. Never mix-feed under 6 months.
9. Baby-Friendly Hospital Initiative (WHO/UNICEF 2018): Ten Steps, including immediate skin-to-skin and initiation within 1 hour, no food or fluids other than breast milk unless medically indicated, 24-hour rooming-in, responsive feeding, and counselling on bottles, teats and pacifiers.
10. Mastitis: keep breastfeeding or expressing from the affected side; flucloxacillin/cloxacillin 500 mg orally 6-hourly for 10–14 days if bacterial; abscess needs drainage.
11. Expressed breast milk: room temperature up to 4 hours, fridge up to 4 days, freezer about 6 months (CDC); feed by cup, not bottle.
12. Formula (only when truly needed): boiled water at ≥ 70 °C, exact scoops as on the tin (usually 1 level scoop per 30 mL), cup feeding, discard any prepared milk not used within 1 hour.

1 Definitions and Key Concepts
2 Burden and Uganda Context
3 Principles and Framework
4 Application in Practice

5 Clinical Acumen
6 Common Problems and Their Management
7 Guidelines, Policy and Law in Uganda
8 Key Points
9 References

1. Definitions and Key Concepts

Infant and young child feeding (IYCF) covers how babies and toddlers are fed from birth to 2 years: **breastfeeding** and **complementary feeding**. It is the cheapest and most powerful child-survival intervention we have.

The rules are short: **breastfeed within 1 hour**, **only breast milk for 6 months**, **add the right foods at 6 months** (thick, energy-dense, diverse, with animal-source foods every day), **keep breastfeeding to 2 years or beyond**, **feed more during and after illness**, and **feed responsively**.

> **CLINICAL PEARL**
> The one thing never to forget: **when a mother says "my milk is not enough", look at a feed before you believe it.** In almost every case the baby is poorly attached or fed too rarely. Fix attachment and frequency and the supply follows. Reaching for formula or "glucose water" is how exclusive breastfeeding ends.

Term	Definition (WHO/UNICEF)
Early initiation of breastfeeding	Putting the newborn to the breast **within 1 hour** of birth
Exclusive breastfeeding (EBF)	Only breast milk (including expressed or wet-nurse milk); **ORS, drops and syrups** of vitamins, minerals or medicines allowed. **No** water, other liquids or foods
Predominant breastfeeding	Breast milk plus water or water-based drinks (not other milks or food)
Mixed feeding	Breast milk plus other milk or foods **before 6 months**
Pre-lacteal feeds	Anything given before breastfeeding starts (sugar water, glucose, herbs, cow's milk): harmful
Complementary feeding	Giving other foods and fluids **in addition to breast milk** from 6 months
Replacement feeding	Feeding a child who receives **no** breast milk with a diet that provides all needs
Breast-milk substitute (BMS)	Any food marketed as a partial or total replacement for breast milk (infant formula, follow-up formula, growing-up milks, and baby bottles and teats under the Code)
Minimum dietary diversity (MDD)	**5 or more of 8 food groups** in the previous day: (1) breast milk, (2) grains, roots, tubers and plantains, (3) pulses, nuts and seeds, (4) dairy, (5) flesh foods (meat, fish, poultry, organ meat), (6) eggs, (7) vitamin A-rich fruits and vegetables, (8) other fruits and vegetables
Minimum meal frequency (MMF)	**2** meals (breastfed 6–8 months), **3** (breastfed 9–23 months), **4** (non-breastfed 6–23 months, including at least 2 milk feeds)

Term	Definition (WHO/UNICEF)
Minimum acceptable diet (MAD)	Meets both MDD and MMF
Responsive feeding	Feeding that responds to the child's hunger and fullness cues, encourages self-feeding and is patient and interactive (WHO 2023)

2. Burden and Uganda Context

- **Global**: optimal breastfeeding alone could prevent about **400,000 under-five deaths a year** worldwide (WHO), mostly from diarrhoea and pneumonia Verify *current WHO figure*. Globally **fewer than one-third (30.8%)** of children aged 6–23 months meet minimum dietary diversity, and only about half get enough meals (WHO fact sheet, 2018–2024 data).
- **Uganda**: about **66%** of babies are put to the breast within 1 hour and about **66%** of infants under 6 months are exclusively breastfed; continued breastfeeding falls to about **43% at 2 years**; only about **15%** of children 6–23 months receive a **minimum acceptable diet** (UDHS 2016, cited in MIYCAN 2021) Verify *whether UDHS 2022 figures supersede these*. Facility data for FY 2022/23 show **92%** initiation within 1 hour among facility births, but home births lag.
- **Stunting (26%, UDHS 2022)** rises steeply from 6 to 24 months: the complementary-feeding window. Poor complementary feeding is the main reason Ugandan children become stunted in this period.
- **Common Ugandan practices that harm**: discarding **colostrum**, **pre-lacteal feeds** (sugar water, glucose, herbs), giving **water or cow's milk** in the first months, **thin porridge** as the main complementary food, sharing a plate with siblings, **stopping breastfeeding abruptly** when the mother becomes pregnant, bottles, and early return to work without support.

> **UGANDA CONTEXT**
> On the ward and in clinic you will mostly meet **grandmothers and older siblings** feeding children while mothers work in the garden or market, so **counsel whoever actually feeds the child**. Maternity leave is short and informal-sector mothers often have none; teach **expressing and cup-feeding** early. Locally available, affordable foods for good complementary feeding include **silverfish (mukene/omena) powder, eggs, beans and groundnut paste (ebinyebwa), sim-sim, dark green leaves (dodo/amaranth, nakati, cassava leaves), orange-fleshed sweet potato, pumpkin, pawpaw, mango, avocado** and **millet or maize porridge enriched with groundnut paste, milk or oil**.

3. Principles and Framework

Physiology of lactation: hormones and reflexes

Hormone	Trigger	Action	Practical lesson
Prolactin (anterior pituitary)	**Suckling**; higher at **night**	Milk **production** for the next feed	Frequent feeds, including at night, make more milk
Oxytocin (posterior pituitary)	Suckling, seeing, hearing or thinking of the baby	**Let-down (milk ejection)**: myoepithelial cells contract; also uterine contraction ("afterpains")	Stress, pain and doubt **inhibit** let-down; a calm, confident mother flows better
Feedback inhibitor of lactation (FIL)	Milk left in the breast	**Reduces** production locally	**Emptying the breast** keeps supply up; engorgement shuts it down

Supply follows demand: the more milk removed, the more is made. Supplements (water, formula) reduce suckling and therefore supply.

What is in breast milk

Component	Features
Colostrum (first 2–4 days, yellow, small volume)	Rich in **secretory IgA, lactoferrin, white cells, vitamin A**, growth factors; laxative (clears meconium, reduces jaundice); "the first immunisation"
Transitional and mature milk	Volume rises to about **500–800 mL/day**; about **65–70 kcal/100 mL**
Foremilk versus hindmilk	Foremilk is watery (quenches thirst); **hindmilk is fat-rich** (energy, satiety). Let the baby finish one breast before offering the other
Protein	Easily digested **whey-dominant** protein; less renal solute load than cow's milk
Fat	Long-chain polyunsaturated fatty acids (DHA, ARA) for brain and retina; **lipase** aids digestion
Carbohydrate	**Lactose** and **human milk oligosaccharides** (feed protective bifidobacteria, block pathogens)
Protective factors	IgA, lactoferrin, lysozyme, macrophages, bifidus factor, anti-inflammatory factors
Micronutrients	**Iron** (low amount but highly absorbed), zinc; adequate vitamin A if the mother is well nourished. **Vitamin K** is low (hence vitamin K at birth); vitamin D low
Water	Enough for a baby in hot weather: **no extra water needed** under 6 months

Why 6 months?

Breast milk meets all energy and nutrient needs for about 6 months. After that, needs for **energy, iron, zinc and vitamin A** exceed what breast milk provides; the infant's gut, kidneys and oral-motor skills (sitting with support, chewing movements, loss of tongue-thrust) are ready. Starting earlier displaces breast milk and adds infection (contaminated food and water); starting later causes **growth faltering** and **iron deficiency**.

The energy gap from 6 to 24 months

Age	Energy needed from complementary foods (breastfed child, about)
6–8 months	**200 kcal/day**
9–11 months	**300 kcal/day**
12–23 months	**550 kcal/day**

A baby's stomach holds about **30 mL/kg**, so foods must be **energy-dense** (thick porridge, with oil, groundnut paste or milk added). Thin, watery porridge fills the stomach with water, not energy.

Benefits of breastfeeding

For the child	For the mother	For family and society
Fewer and less severe **diarrhoea, pneumonia, otitis media, sepsis**; less NEC in preterms	**Less PPH** (oxytocin), faster uterine involution	Free, safe, always available
Lower infant mortality, including SIDS	**Lactational amenorrhoea** (LAM, > 98% protection when all 3 criteria met)	Less spending on formula and health care
Higher IQ scores	Lower risk of **breast and ovarian cancer** and **type 2 diabetes**	Environmentally clean
Less **obesity and diabetes** later	Bonding, convenience	

WHO 2023 complementary feeding recommendations (6–23 months)

1. **Continue breastfeeding to 2 years or beyond** (strong).
2. **Milks** for non-breastfed children: **6–11 months**: infant formula **or** (full-fat) animal milk; **12–23 months**: **animal milk**; **follow-up and growing-up formulas are not recommended**; no flavoured or sweetened milks.
3. **Introduce complementary foods at 6 months (180 days)** while breastfeeding continues (strong).
4. **Dietary diversity**: **animal-source foods (meat, fish, eggs) daily**; **fruits and vegetables daily**; **pulses, nuts and seeds frequently**; minimise starchy staples; prefer whole grains.
5. **Unhealthy foods**: no foods high in **sugar, salt or trans fats**; **no sugar-sweetened beverages**; **no non-sugar sweeteners**; **limit 100% fruit juice**.
6. **Supplements and fortified foods** where needs cannot be met (micronutrient powders, fortified cereals, small-quantity LNS) as part of counselling, not instead of a diverse diet.
7. **Responsive feeding** (strong).

The six dimensions of good complementary feeding ("FATVAH"): Frequency, Amount, Thickness (density), Variety, Active (responsive) feeding, Hygiene.

4. Application in Practice

For a feeding topic, the workup is the **feeding assessment and counselling session** (IMCI "assess the child's feeding" and "counsel the mother").

Step 0: first 5 minutes

1. **Check general danger signs** (IMCI): unable to drink or breastfeed, vomits everything, convulsions, lethargic or unconscious. Any → treat and refer; feeding counselling can wait.
2. **Young infant** (< 2 months): temperature, jaundice, hypoglycaemia risk; **weigh**.
3. **Classify nutrition**: weight, length or height, **MUAC** (≥ 6 months), **oedema**. SAM → nutrition pathway (see the Malnutrition topic).
4. Create **privacy and a comfortable seat** for the mother; introduce yourself; ask permission to watch a feed.

Step 1: focused feeding history

Ask	Why it matters
When was the baby first put to the breast? Pre-lacteal feeds?	Delayed initiation and pre-lacteals predict early stopping and infection
Breastfeeding now? How often by day and **by night**? How long? One or both breasts?	Frequency drives supply; < 8 feeds in 24 h is too few for a young infant
Anything else given (water, glucose, cow's milk, formula, herbs, porridge)? From what age?	Mixed feeding before 6 months: infection risk, lower supply, HIV transmission risk
Feeding utensils: cup, bottle, spoon? How cleaned?	Bottles carry infection and interfere with suckling; switch to a cup
Wet nappies, stools, weight gain on the card	Adequacy of intake
Breast problems: pain, cracked nipples, lumps, fever	Attachment problems, mastitis, thrush, engorgement
6–23 months: 24-hour recall: what, how much, how thick, how many meals and snacks, **animal-source foods**, vegetables and fruit, oil	Energy density, diversity (MDD), frequency (MMF)
Who feeds the child? Own plate? Helped to eat?	Responsive feeding; sibling competition; counsel the real feeder

Ask	Why it matters
Feeding during illness	Many families withhold food during diarrhoea
Mother's HIV status, ART, viral load; child's EID results	Feeding duration, infant prophylaxis, testing schedule
Mother's health: nutrition, mood (depression), work, contraception, new pregnancy	Support needs; abrupt weaning risk
Immunisation, vitamin A, deworming	Missed prevention opportunities

Step 2: examination (Hutchison order, adapted)

1. **Introduction and consent**: greet, sit at the mother's level, ask permission; privacy; **hand hygiene**.
2. **Observe a breastfeed** first (at least a few minutes), before touching: **positioning** (4 points) and **attachment** (4 signs), **effective suckling** (slow deep sucks with pauses, **audible swallowing**, cheeks round not drawn in), baby releases the breast and looks satisfied.
3. **Baby, general inspection**: alert or lethargic, hydration, jaundice, wasting; weight, length, head circumference, **plot**.
4. **Baby, face and mouth**: **oral thrush**, cleft palate, **tongue-tie**; then tone and alertness; signs of illness.
5. **Mother's breasts** (with consent and privacy, chaperone if needed): engorgement, nipple shape (flat or inverted), **fissures**, redness, tender lump (blocked duct, mastitis), fluctuant mass (abscess), thrush.
6. **Older child**: anthropometry (WHZ, HAZ, MUAC), oedema, pallor, development (can sit? self-feed?), and **watch a meal** if possible.
7. **Thank** the mother, **wash hands**, **summarise** and agree 2–3 actions.

Document these positives	And these negatives
"Observed feed: good positioning, 4/4 attachment signs, audible swallowing"	"No nipple fissures, no breast lump or redness"
Weight and Z-scores; trend on the card	"No oral thrush; no tongue-tie; palate intact"
Number of feeds and wet nappies in 24 h	"No other fluids given"
24-h recall: food groups and meals	"No oedema; MUAC 13.8 cm"

Step 3: bedside tests

Test	How	What it tells you
Weight on a calibrated scale and plotting	Undressed baby	Adequacy of intake; faltering
Wet nappy count	Ask; check a nappy	≥ 6/day after day 5 = adequate fluid
Observed breastfeed (BFHI/IMCI form)	Watch 3–5 min	Attachment, suckling, positioning: the commonest fixable cause
24-hour recall	Ask the feeder	MDD, MMF; specific counselling
MUAC and oedema (≥ 6 months)	Tape, press feet	Acute malnutrition
Random blood glucose in a poorly feeding young infant	Glucometer	Hypoglycaemia
Bilirubin (transcutaneous or serum) if jaundiced	Meter or laboratory	Breastfeeding versus breast-milk jaundice, pathological causes

Test	How	What it tells you
HIV rapid test for the mother (retesting) or **DNA PCR** for the exposed infant	Per schedule	Feeding plan and prophylaxis

Step 4: laboratory and imaging

Test	When	Expected or abnormal finding	How it changes management
Serum sodium	Poorly feeding, jaundiced, lethargic newborn with weight loss > 10%	**Hypernatraemia** (> 150 mmol/L)	Breastfeeding-failure dehydration: slow rehydration, intensive lactation support, temporary supplementation with EBM or formula. RRH and NRH
Hb (HemoCue)	Pallor, 6–24 months with poor diet	Anaemia (iron deficiency)	Iron-rich foods, iron supplement, deworm. HC III and above
Maternal viral load	HIV-positive breastfeeding mother: **every 3 months until breastfeeding ends** (Uganda; see HIV in Pregnancy topic)	Non-suppressed (≥ 1000 copies/mL)	Adherence support, extended infant prophylaxis, enhanced EID
Infant DNA PCR	HIV-exposed: 4–6 weeks, 9 months, 6 weeks after stopping breastfeeding; rapid test at 18 months	Positive	Start ART; continue breastfeeding to 24 months
Breast ultrasound or aspiration	Suspected abscess	Fluid collection	Needle aspiration or incision and drainage
Pus culture	Recurrent or non-responding mastitis	*S. aureus* (MRSA)	Targeted antibiotic

Step 5: supporting breastfeeding from birth (labour ward and postnatal ward)

1. **Immediate skin-to-skin** after birth (vaginal or CS) for at least 1 hour; **initiate breastfeeding within 1 hour**; delay weighing and bathing.
2. **Colostrum is precious**: no pre-lacteal feeds.
3. **Rooming-in 24 hours**; **feed on demand** (responsive feeding), at least **8 times in 24 hours**, including nights; watch for early hunger cues (rooting, sucking fingers, moving lips) rather than waiting for crying.
4. Help with **positioning and attachment** at the first feeds:

Positioning (4 points)	Attachment (4 signs)
Baby's **head and body in a straight line**	**More areola visible above** the baby's mouth than below
Baby held **close** to the mother's body	**Mouth wide open**
Baby **facing the breast**, nose opposite the nipple	**Lower lip turned outward**
Whole body **supported** (not just neck and shoulders)	**Chin touching the breast**

Effective suckling: slow deep sucks with pauses, swallowing heard, cheeks rounded. Positions: cradle, cross-cradle, football (rugby) hold (good after CS or for twins), lying down.

5. **After CS**: skin-to-skin in theatre or recovery as soon as the mother is responsive; side-lying or football hold; help with lifting.
6. **Small and preterm babies**: **kangaroo mother care**, expressed breast milk by **cup, spoon or NG tube**, then breast as suck matures (see Neonatal topics).

7. **Teach expressing by hand** to every mother before discharge (thumb above, fingers below the areola, press back toward the chest, compress, release; rotate around the breast).
8. **Contraception**: LAM, progestogen-only methods, IUD; **avoid combined pills in the first 6 weeks** while breastfeeding (see Puerperium topic).

Step 6: exclusive breastfeeding 0–6 months, counselling messages

1. **Breast milk is all your baby needs**, even in hot weather; no water, glucose, herbs, juice or porridge.
2. Feed **whenever the baby wants**, day and night, **at least 8 times** in 24 hours; let the baby finish one breast (hindmilk) before offering the other.
3. Signs your baby is getting enough: **6 or more wet nappies a day**, regular stools, **gaining weight** on the card, content after feeds.
4. **Going back to work**: express milk before leaving; the caregiver gives it by **cup**; breastfeed more at night and on days off. Storage (CDC): **room temperature up to 4 hours**, **fridge up to 4 days**, **freezer about 6 months**; thawed milk used within 24 h (fridge) and within 2 h once warmed; never refreeze.
5. **Medicines and illness**: most maternal illnesses and medicines (including ART, antimalarials, antibiotics, TB drugs) are compatible with breastfeeding; continue breastfeeding through the mother's and baby's illnesses.

Step 7: complementary feeding from 6 months (UCG 2023 IMCI, 250 mL cup)

Age	Texture	Frequency	Amount per meal
6–9 months	Thick porridge; well-mashed foods	**2–3 meals** + **1–2 snacks**	Start with **2–3 tablespoons**, increase to **½ cup**
9–12 months	Finely chopped or mashed family food; finger foods	**3–4 meals** + **1–2 snacks**	**½ cup**
12–24 months	Family foods, chopped if needed	**3–4 meals** + **1–2 snacks**	**¾ cup**
2 years and over	Family foods	**3–4 meals** + **1–2 snacks**	**At least 1 full cup**
Not breastfed (6–23 months)	As above	**Add 1–2 extra meals** and **2 or more milk feeds** daily (MMF 4)	

Snacks in UCG: egg, banana, bread, avocado. UCG also says give a nutritious snack 3 times a day if breastfed and 5 times a day if not breastfed at 6–9 months.

Making a good Ugandan complementary meal ("enrich the porridge")

- **Thick** enough to stay on a spoon.
- Add **energy**: a teaspoon of **oil**, **groundnut or sim-sim paste**, **milk**.
- Add **protein and iron or zinc**: **egg**, **silverfish (mukene) powder**, **liver**, meat, fish, **beans** (mashed).
- Add **vitamin A-rich** foods: **orange sweet potato, pumpkin, carrots, pawpaw, mango, dark green leaves**.
- Use **iodised salt** in small amounts only for older children (no added salt is ideal under 1 year); **no sugar-sweetened drinks or sweets**.

Hygiene: wash hands (caregiver and child) with soap; clean utensils; feed with a **cup and spoon** (no bottles); serve food immediately after cooking or store it safely; boil or treat drinking water.

Responsive feeding: sit with the child, give **their own bowl**, encourage but do not force, talk and make eye contact, offer new foods repeatedly (several "tastes"), minimise distractions.

Step 8: feeding during and after illness

1. **Breastfeed more often and for longer** at each feed.
2. Offer **extra fluids** (ORS in diarrhoea; see Diarrhoea topic) and **soft, favourite foods**, small amounts often.

3. **Never withhold food** during diarrhoea or fever.
4. After illness: **an extra meal a day for at least 2 weeks** (catch-up).
5. Sick HIV-exposed child: at least **2 extra meals and 1 snack** (MIYCAN 2021).

Step 9: micronutrients and prevention around feeding

Intervention	Schedule (Uganda)
Vitamin A	**100,000 IU at 6–11 months**; **200,000 IU every 6 months from 12 to 59 months**
Deworming	From **1 year, every 6 months**: albendazole **200 mg (12–23 months)**, **400 mg (≥ 2 years)** (or mebendazole 500 mg)
Iron	Iron-rich foods; supplements where anaemia is common or with low birth weight (per programme)
Iodised salt	Household level
Vitamin K	At birth (newborn)
Immunisation	UNEPI schedule; every feeding contact = vaccination check

Putting it together

Model one-line summary

> "A 7-week-old girl, weight 3.9 kg (birth weight 3.2 kg; WAZ −1.6, crossing down one line), breastfed 6 times a day plus water and glucose 'because of the heat', mother with cracked nipples from poor attachment, HIV-negative mother: **feeding problem (mixed feeding, infrequent feeds, poor attachment) with slow weight gain**, no danger signs."

Problems: (1) poor attachment causing sore nipples and ineffective milk transfer; (2) infrequent feeds; (3) water and glucose (mixed feeding); (4) slow weight gain.

Plan: show positioning and attachment, feed **8–12 times** in 24 h including nights, **stop water and glucose**, nipple care (expressed milk on the nipple, air-dry), reassure, **review weight in 5–7 days** (IMCI: follow up a feeding problem in 5 days, low weight in 14 days).

Model case presentation (ward round)

> "Baby Kato is a **5-month-old boy**, **birth weight 3.0 kg** at term by SVD at a HC IV. He was put to the breast **2 hours after birth** after being given **glucose water**. He is breastfed about **6 times a day**, and since 3 months has also had **water, gripe water and diluted cow's milk** by bottle because his mother returned to her market stall. He is **HIV-exposed**: mother on TLD since ANC, **last viral load 4 months ago < 50 copies/mL**; he completed NVP prophylaxis and had a **negative DNA PCR at 6 weeks**. Immunisations are **up to date to 14 weeks**. Two episodes of diarrhoea in the last month. **Weight 5.6 kg (WAZ −2.2), length 63 cm (LAZ −1.3, WLZ −2.0)**; his card shows the line **flattening since 3 months**. Alert, no danger signs, no thrush. Observed feed: baby's body turned away, **chin off the breast, more areola below than above**, quick shallow sucks. **Summary**: **mixed feeding with poor attachment and bottle use in an HIV-exposed infant**, with **growth faltering** (borderline moderate wasting) and recurrent diarrhoea. **Plan**: correct positioning and attachment, **exclusive breastfeeding** (stop water, cow's milk and bottle), **express breast milk** for the caregiver to give by **cup** while the mother is at the market, feed on demand day and night; maternal **viral load now**; **review in 5 days** and weigh again in 14 days; prepare for **complementary feeding at 6 months**; next **DNA PCR at 9 months**."

5. Clinical Acumen

Red flags

RED FLAG

A baby who is not feeding well is sick until proven otherwise ("unable to drink or breastfeed" is an IMCI danger sign). Also: **weight loss > 10% of birth weight** in the first week, not back to birth weight by **2 weeks**, **fewer than 6 wet nappies** after day 5, **lethargy or jaundice** with poor feeding (hypernatraemic dehydration, sepsis), a **flat or falling growth line**, a mother with **fever and a hot, red breast** (mastitis or abscess), and a **depressed or overwhelmed mother**.

Thinking like a clinician

- **"Not enough milk" is usually poor attachment or infrequent feeding**, sometimes pain, a sleepy jaundiced baby, or a separated mother–baby pair. True primary low supply is rare.
- **Crying is not proof of hunger.** Check weight gain and wet nappies before suggesting supplements.
- **Sore nipples mean poor attachment** until proven otherwise (or thrush if painful after weeks of comfortable feeding).
- **A breastfed baby with frequent loose yellow stools and good weight gain is normal**, not diarrhoea.
- **At 6 months, ask "what is in the bowl?"**: thin porridge alone will not do; add oil, groundnut paste, milk, egg or fish powder.
- **The HIV question changes duration, not the method**: exclusive breastfeeding on maternal ART is the safe choice in Uganda.

Common student mistakes

- Advising water for a hot baby under 6 months.
- Telling a mother with mastitis to stop breastfeeding.
- Recommending formula to an HIV-positive mother.
- Giving the age-blind advice "feed porridge three times a day" without quantities, density or diversity.
- Forgetting to ask who feeds the child.

Differential diagnosis of "poor feeding" and "faltering growth" in infancy

Condition	Distinguishing features	Key investigation
Feeding or attachment problem	Sore nipples, short or infrequent feeds, unsettled baby, otherwise well	Observed feed
Inadequate complementary diet	Faltering after 6 months, thin porridge, few food groups	24-hour recall
Neonatal sepsis or meningitis	Sudden poor feeding, fever or hypothermia, lethargy, apnoea	FBC, blood culture, LP
Neonatal jaundice (sleepy, poor suck)	Yellow skin, especially day 1 (pathological) or with dehydration	Bilirubin
Hypernatraemic dehydration	Weight loss > 10%, irritability, fever, few wet nappies	Sodium
Oral problems: thrush, cleft palate, tongue-tie	Visible on inspection; nasal regurgitation with cleft	Examination
Congenital heart disease	Sweating and breathlessness with feeds, murmur	Echo
Neurological disorder (cerebral palsy, birth asphyxia)	Poor suck and swallow, abnormal tone	Neurological examination

Condition	Distinguishing features	Key investigation
HIV infection	Faltering, thrush, recurrent infections, lymphadenopathy	DNA PCR
TB	Faltering, cough, contact	CXR, Xpert
Recurrent infections (diarrhoea, malaria, worms)	History of repeated illness	Malaria test, stool
Gastro-oesophageal reflux	Effortless vomiting after feeds, thriving	Clinical
Pyloric stenosis	Projectile non-bilious vomiting at 2–8 weeks, hungry	Ultrasound, electrolytes
Maternal depression or neglect	Withdrawn mother, missed visits	Screen (EPDS)

Scenario

EXAM FAVOURITE

Worked example: a 24-hour recall in a 10-month-old. Yesterday: breastfed 4 times; **thin maize porridge** (cup, morning); **matooke with bean sauce** (½ cup, evening). Food groups: breast milk (yes), grains, roots and plantains (yes), pulses (yes) = **3 of 8** (MDD needs 5). Meals: 2 (MMF for a breastfed 9–23-month-old is 3). **Neither MDD nor MMF → no minimum acceptable diet.** Counsel: **3–4 meals + 1–2 snacks**, **½ cup per meal** (250 mL cup), make porridge **thick** and add **groundnut paste, milk or oil**; add **an animal-source food daily** (egg, silverfish powder, liver, milk), **dark green leaves and an orange fruit or vegetable daily** (dodo, pawpaw, mango, orange sweet potato). Snacks: banana, avocado, boiled egg. Give her **her own bowl** and help her eat.

6. Common Problems and Their Management

Breastfeeding problems

Problem	Features	Management
"Not enough milk" (perceived)	Crying, frequent feeds, mother's doubt; weight gain often normal	Observe a feed, check weight and wet nappies, **improve attachment**, **feed more often** (including nights), stop supplements, build confidence; review in 5–7 days
True low intake	Poor weight gain, < 6 wet nappies	Correct attachment; express after feeds to increase supply; skin-to-skin; temporary supplement with **expressed milk by cup**; treat causes (pain, separation, sleepy jaundiced baby)
Sore or cracked nipples	Pain during feeds, fissures	**Correct attachment**; do not stop feeding; start on the less sore side; rub **hindmilk** on the nipple and air-dry; avoid soap and creams; if very painful, express from that side for 1–2 days and cup-feed
Engorgement	Both breasts full, hard, painful, shiny; milk not flowing	**Frequent feeding** with good attachment; **hand-express** a little to soften the areola before feeding; **warm compress before**, **cold compress after**; paracetamol; support bra
Blocked duct	Tender lump, mother well	Continue feeding (start on that side), gentle massage toward the nipple, vary positions, avoid tight clothing
Mastitis	Red, hot, tender wedge, **fever, malaise**	**Continue breastfeeding or expressing** from the affected side (essential); rest, fluids, **paracetamol or ibuprofen**; if bacterial (fissure, no improvement in 24 h, very ill): **flucloxacillin or cloxacillin 500 mg orally 6-hourly for 10–14 days** (erythromycin if penicillin-allergic)
Breast abscess	Fluctuant, very tender swelling, fever	**Needle aspiration or incision and drainage**, antibiotics, continue feeding from the other breast and resume the affected side once drained

Problem	Features	Management
Flat or inverted nipples	Nipple does not protrude	Good attachment matters more than nipple shape (the baby suckles the areola); skin-to-skin; try positions; draw out with a syringe before feeds
Thrush (candida)	Burning nipple pain after weeks of comfortable feeds; white patches in the baby's mouth	Treat **both**: **nystatin oral suspension** for the baby (1 mL 4 times daily for 7 days) and **miconazole cream** or nystatin to the nipples
Breastfeeding (suboptimal intake) jaundice	First week, poor feeding, weight loss	More frequent effective feeds; treat dehydration; phototherapy per thresholds
Breast-milk jaundice	Well, thriving baby, jaundice beyond 2 weeks	Exclude pathological causes (conjugated bilirubin, thyroid, infection); **continue breastfeeding**
Low birth weight or preterm	Weak suck	KMC, EBM by cup or NG, breastfeed as soon as able
Twins	Both can be exclusively breastfed	Football hold, feed together or one after the other
Cleft lip or palate	Difficulty forming a seal; nasal regurgitation	Upright position, EBM by cup or special feeder; refer for surgery (Uganda: CoRSU, Mulago)
Mother separated or working	Missed feeds	Expressing, cup feeding by the caregiver, night breastfeeding

Drug doses used in this topic

Drug	Dose	Route	Frequency	Duration	Key cautions
Flucloxacillin or cloxacillin (mother)	**500 mg**	Oral	6-hourly	**10–14 days**	Bacterial mastitis; erythromycin if penicillin-allergic; continue breastfeeding
Paracetamol (mother)	1 g	Oral	6-hourly (max 4 g/day)	As needed	Engorgement, mastitis pain
Nystatin oral suspension (baby)	**1 mL**	Oral	4 times daily	**7 days**	Treat mother's nipples at the same time
Miconazole cream (mother)	Thin layer to nipples	Topical	After feeds	Until healed	With treatment of the baby
Vitamin A	**100,000 IU** (6–11 months); **200,000 IU** (12–59 months)	Oral	Every 6 months	Routine	Do not repeat within 1 month
Albendazole	**200 mg** (12–23 months); **400 mg** (≥ 2 years)	Oral	Every 6 months	Single dose	Or mebendazole 500 mg

Feeding the HIV-exposed infant

Principle: with the mother on effective **ART (TLD)** and **virally suppressed**, the risk of transmission through breastfeeding is very low, while not breastfeeding greatly increases death from diarrhoea, pneumonia and malnutrition in Uganda. So **breastfeeding is recommended for all women irrespective of HIV status** (MIYCAN

2021).

1. **Mother**: lifelong **ART**, **viral load monitoring** during breastfeeding (every 3 months until breastfeeding ends, per the HIV-in-pregnancy topic), adherence support, healthy breasts (treat mastitis and cracked nipples promptly; they increase transmission).
2. **Infant**: risk-based **prophylaxis** (NVP, or AZT + NVP if high risk), **co-trimoxazole** from 6 weeks, **EID**: DNA PCR at **4–6 weeks**, **9 months**, and **6 weeks after complete cessation** of breastfeeding; rapid test at **18 months**.
3. **Feeding**:
   - **Exclusive breastfeeding for 6 months**; **no mixed feeding** under 6 months (gut inflammation lets virus cross).
   - From 6 months: **complementary foods with continued breastfeeding**.
   - **Duration**: **WHO 2016**: breastfeed for **at least 12 months**, and may continue **up to 24 months or longer** (like the general population) with full ART adherence support. **Uganda MIYCAN 2021**: for **HIV-negative exposed infants**, encourage **stopping at 12 months** and give **at least 500 mL/day of other milk** (cow, goat or soya milk); for **HIV-positive infants**, **continue breastfeeding on demand to 24 months**.
   - **Stopping**: gradually over **1–2 weeks** (MIYCAN): express and cup-feed, gradually replace with other milk; never abrupt. (UCG advises about 1 month for stopping breastfeeding in general.)
   - **Replacement feeding** only if breastfeeding is impossible and it is **acceptable, feasible, affordable, sustainable and safe**, with a guaranteed supply and safe preparation; never combine with breastfeeding.

> UGANDA CONTEXT
> The **UCG 2023 HIV chapter still contains older wording** ("exclusive breastfeeding stopping at 3–6 months if replacement feeding is possible"). Current Ugandan practice follows the **consolidated HIV guidelines and MIYCAN 2021**: **exclusive breastfeeding for 6 months on ART, continued breastfeeding with complementary foods**, stopping at about **12 months** for HIV-negative infants (WHO allows up to 24 months), and **24 months** if the infant is HIV-positive. Do not advise formula as the default for an HIV-positive mother.

When is breastfeeding not advised?

Very few situations (WHO acceptable medical reasons; CDC):

- **Infant**: **classic galactosaemia**; some other rare metabolic diseases (for example maple syrup urine disease, phenylketonuria: partial breastfeeding with special formula).
- **Mother**: **Ebola virus disease** (in Uganda's outbreaks: separate and give replacement feeds); certain **drugs** (cytotoxic chemotherapy, radioactive iodine, some radiopharmaceuticals; check each drug); **herpes simplex lesions on the breast** (avoid that breast until healed); severe maternal illness preventing care (express if possible).
- **Not contraindications in Uganda**: **HIV on ART**, **hepatitis B** (baby gets birth-dose vaccine), **TB** (baby gets TB preventive therapy; mother on treatment can breastfeed), **malaria**, mastitis, most antibiotics and antimalarials, **ART**.

When formula is truly needed (safe preparation)

1. Use **commercial infant formula** under 12 months (WHO 2023 allows full-fat animal milk from 6 months for non-breastfed infants).
2. **Wash hands**; **boil water** and use it while still at least **70 °C** (kills *Cronobacter*), then cool.
3. **Measure exactly**: usually **1 level scoop per 30 mL** of water (follow the tin); **never add extra scoops or sugar** (hypernatraemia, obesity) and **never dilute** (malnutrition).
4. **Feed by cup**, not bottle; **discard** any prepared milk not used within **1 hour** (UCG).
5. Approximate volume: **150 mL/kg/day** for young infants.

6. Teach and **observe** preparation; ensure a **guaranteed supply** (an interrupted formula supply is dangerous).

Consequences of suboptimal feeding

Group	Consequences
Infant (0–6 months)	Diarrhoea, pneumonia, otitis media, sepsis and death (mixed or no breastfeeding); hypernatraemic dehydration and jaundice (insufficient intake); higher HIV transmission with mixed feeding
Young child (6–24 months)	**Growth faltering, stunting, wasting**, micronutrient deficiencies (**iron-deficiency anaemia**, vitamin A deficiency, zinc deficiency), developmental delay
Long term	Lower school achievement and income; overweight, obesity and diabetes (especially with formula and sugary foods)
Mother	Engorgement, mastitis, abscess, early return of fertility, higher risk of breast and ovarian cancer and type 2 diabetes when not breastfeeding
Formula-related	Contamination and diarrhoea, over-concentration (hypernatraemia), over-dilution (malnutrition), cost

7. Guidelines, Policy and Law in Uganda

- **MoH Maternal, Infant, Young Child and Adolescent Nutrition (MIYCAN)** guidelines and training (2021).
- **UCG 2023**: IYCF (19.1.1), IMCI feeding recommendations (17.3.12), HIV infant feeding.
- **BFHI** in maternity units; growth monitoring and promotion through the Child Health Card.
- **WHO 2023** complementary feeding guideline and **WHO 2016** HIV and infant feeding update.

The Baby-Friendly Hospital Initiative: Ten Steps (WHO/UNICEF 2018)

Step	Content
1a	Comply fully with the **International Code of Marketing of Breast-milk Substitutes** and World Health Assembly resolutions
1b	Have a **written infant feeding policy** routinely communicated to staff and parents
1c	Establish ongoing **monitoring and data-management** systems
2	Ensure staff have sufficient **knowledge, competence and skills** to support breastfeeding
3	Discuss the importance and management of breastfeeding with **pregnant women and their families**
4	Facilitate **immediate and uninterrupted skin-to-skin** contact and support mothers to **initiate breastfeeding as soon as possible** after birth
5	Support mothers to **initiate and maintain** breastfeeding and **manage common difficulties**
6	Do not provide breastfed newborns any **food or fluids other than breast milk**, unless **medically indicated**
7	Enable mothers and infants to remain together and practise **rooming-in 24 hours a day**
8	Support mothers to recognise and respond to their infants' **cues for feeding**
9	Counsel mothers on the use and risks of **feeding bottles, teats and pacifiers**
10	Coordinate **discharge** so that parents and infants have timely access to **ongoing support and care**

Steps 1–2 are "critical management procedures"; steps 3–10 are "key clinical practices". The Code also bans advertising of BMS to the public, free samples to mothers, and gifts to health workers.

UGANDA CONTEXT

Uganda enforces the Code through its **Regulations on Marketing of Infant and Young Child Foods**: no free formula samples or promotion in health facilities, labels must carry warnings and preparation instructions, and facilities must **not accept donations of formula or complementary foods** from manufacturers, including during emergencies (MIYCAN 2021). As a student you must not accept gifts or sponsored materials from BMS companies.

Roles by level of care

Level	Role
Community / VHT	Promote early initiation and EBF; home visits in the first week; counsel on complementary feeding; MUAC screening; mother support groups
HC II–III	IMCI feeding assessment and counselling; growth monitoring and promotion; vitamin A and deworming; HIV testing and EID; manage common breast problems
HC IV / hospital	**BFHI** practices on maternity and postnatal wards; lactation support for sick and small babies (KMC units); breast abscess drainage; specialist IYCF counselling for HIV-exposed infants; nutrition units

8. Key Points

High-yield points

- **1-6-24**: breastfeed within 1 hour, exclusively for 6 months, continue to 24 months or beyond.
- **Observe a feed before believing "not enough milk"**; fix attachment and frequency.
- **EBF allows ORS, drops and syrups only**: no water, even in hot weather.
- **4 positioning points and 4 attachment signs.**
- **At 6 months, ask "what is in the bowl?"**: thick, enriched, diverse, animal-source food daily.
- **MDD ≥ 5 of 8 food groups; MMF 2/3/4**; only about 15% of Ugandan children get a minimum acceptable diet.
- **Feed more during and after illness**; never withhold food.
- **HIV on ART: breastfeed exclusively for 6 months, never mix-feed**; duration per MIYCAN.
- **Mastitis: keep milk moving.**
- **Formula only when truly needed**, prepared with water ≥ 70 °C, by cup, discarded after 1 hour.

Numbers & doses

Item	Value
Initiation	Within **1 hour** of birth
Exclusive breastfeeding	**6 months** (180 days)
Continued breastfeeding	To **2 years or beyond**
Feeds per day (young infant)	**8–12** in 24 h, including nights
Adequate intake	**≥ 6 wet nappies/day** after day 5; birth weight regained by **10–14 days**
Milk volume and energy	**500–800 mL/day**; about **65–70 kcal/100 mL**
Energy from complementary foods	**200 / 300 / 550 kcal/day** at 6–8 / 9–11 / 12–23 months
Infant stomach capacity	About **30 mL/kg**
Meals (UCG/IMCI)	6–9 mo **2–3 + 1–2 snacks**; 9–24 mo **3–4 + 1–2 snacks**

Item	Value
Amount per meal (250 mL cup)	6–9 mo **2–3 tbsp → ½ cup**; 9–12 mo **½ cup**; 12–24 mo **¾ cup**; ≥ 2 y **1 cup**
MDD	**≥ 5 of 8** food groups
MMF	**2** (breastfed 6–8 mo), **3** (breastfed 9–23 mo), **4** (non-breastfed, including 2 milk feeds)
Uganda MAD	About **15%** (UDHS 2016)
Uganda EBF / early initiation	About **66% / 66%** (UDHS 2016)
HIV-exposed, HIV-negative infant (Uganda MIYCAN)	Stop at **12 months**, then **≥ 500 mL/day** other milk; WHO: at least 12, up to 24 months
HIV-positive infant	Breastfeed to **24 months**
EID final test	DNA PCR **6 weeks after stopping** breastfeeding
Stopping breastfeeding (HIV)	Gradually over **1–2 weeks** (MIYCAN)
Mastitis antibiotic	**Flucloxacillin or cloxacillin 500 mg 6-hourly × 10–14 days**
Nystatin (baby)	**1 mL 4 times daily × 7 days**
EBM storage (CDC)	**4 h** room, **4 days** fridge, **~6 months** freezer
Formula preparation	Water **≥ 70 °C**; scoops exactly as on the tin; discard after **1 hour**; about **150 mL/kg/day**
Vitamin A	**100,000 IU** 6–11 mo; **200,000 IU** 6-monthly 12–59 mo
Deworming	From 1 year 6-monthly: albendazole **200 mg** (< 2 y) / **400 mg**
LAM protection	**> 98%** when all 3 criteria are met

Exam traps

- **"Give water to a breastfed baby in hot weather."** Wrong: breast milk contains enough water under 6 months.
- **"Stop breastfeeding because of mastitis."** Wrong: keep milk moving; it is the treatment.
- **"Formula is safer for babies of HIV-positive mothers."** Wrong in Uganda: EBF on maternal ART is recommended.
- **"Introduce porridge at 4 months because the baby cries."** Wrong: complementary foods at 6 months.
- **"Thin porridge three times a day is adequate at 9 months."** Wrong: thick, enriched, diverse foods, 3–4 meals plus snacks.
- **"Follow-up formula for a non-breastfed 15-month-old."** Wrong (WHO 2023): animal milk; follow-up formulas not recommended.
- **"Sore nipples are normal and will settle."** Wrong: they signal poor attachment.
- **"Mix-feed while the HIV-positive mother returns to work."** Wrong: exclusive breastfeeding with expressed milk by cup.

Questions seniors ask

Viva questions

1. *What does exclusive breastfeeding allow?* Breast milk only, plus ORS, drops or syrups of vitamins, minerals and medicines.
2. *Name the 4 signs of good attachment.* More areola above than below, mouth wide open, lower lip turned out, chin touching the breast.

3. *Why start complementary foods at 6 months?* Breast milk no longer meets energy, iron, zinc and vitamin A needs; the infant is developmentally ready; earlier starts add infection and displace breast milk.
4. *What does WHO 2023 say about milks for non-breastfed children?* 6–11 months: formula or animal milk; 12–23 months: animal milk; follow-up formulas not recommended.
5. *How should an HIV-positive mother in Uganda feed her baby?* On ART with viral load monitoring: EBF for 6 months, then complementary foods with continued breastfeeding (to 12 months if the infant is HIV-negative per MIYCAN; up to 24 months per WHO; 24 months if the infant is HIV-positive); no mixed feeding; stop gradually; EID 6 weeks after stopping.
6. *A mother with mastitis asks if she should stop breastfeeding.* No: continued emptying of the breast is the treatment; add analgesia and flucloxacillin or cloxacillin if bacterial.
7. *List the BFHI Ten Steps' key clinical practices.* Antenatal counselling, immediate skin-to-skin and early initiation, support to breastfeed and manage problems, no supplements unless medically indicated, rooming-in, responsive feeding, counselling on bottles, teats and pacifiers, coordinated discharge.

On the ward round

- **This mother says she has no milk on day 2 after CS. What do you do?** Explain that colostrum comes in small amounts and is enough; put the baby skin-to-skin and to the breast often (8–12 times), help with the football hold, hand-express colostrum and give by cup or spoon if needed; check the baby's weight, output and jaundice; avoid formula unless medically indicated.
- **How do you assess a breastfeed?** Watch for the 4 positioning points (in line, close, facing, supported), 4 attachment signs (more areola above, wide mouth, lower lip out, chin to breast) and effective suckling (slow deep sucks with pauses and swallowing).
- **What foods would you recommend for a 9-month-old in rural Uganda?** Breastfeed on demand; 3–4 meals of ½ cup plus 1–2 snacks; thick millet or maize porridge enriched with groundnut paste, milk or oil; mashed matooke or sweet potato with beans and silverfish powder or egg; dark green leaves; pawpaw, mango or banana; own bowl; clean hands and utensils.
- **How long should an HIV-positive mother breastfeed?** On ART with suppressed viral load: EBF for 6 months, then complementary foods with continued breastfeeding; Uganda MIYCAN 2021 advises stopping at 12 months if the infant is HIV-negative (WHO allows up to 24 months), and continuing to 24 months if the infant is HIV-positive; stop gradually over 1–2 weeks; PCR 6 weeks later.
- **What are the BFHI Ten Steps, and which one do Ugandan wards break most?** Policy, staff competence, antenatal counselling, immediate skin-to-skin and early initiation, support, no unnecessary supplements, rooming-in, responsive feeding, counselling on bottles and teats, discharge support. Common breaches: delayed initiation after CS, pre-lacteal glucose water, and separating mother and baby.
- **A breastfed 6-day-old has jaundice and has lost 12% of birth weight. What are you worried about?** Suboptimal intake (breastfeeding failure) with hypernatraemic dehydration and hyperbilirubinaemia; check sodium and bilirubin, observe a feed, rehydrate, support lactation and supplement with expressed milk (or formula if needed) by cup, and look for sepsis.

Mnemonics

- **"1-6-24"**: breastfeed within **1 hour**, exclusively for **6 months**, continue to **24 months** or beyond.
- **Attachment "ALLC"**: **A**reola more above, **L**ips (lower) turned out, **L**arge (wide) open mouth, **C**hin touching.
- **Positioning "In line, Close, Facing, Supported"**.
- **Complementary feeding "FATVAH"** (WHO): **F**requency, **A**mount, **T**hickness (density), **V**ariety, **A**ctive or responsive feeding, **H**ygiene.
- **"2-3-4"** meals: 2–3 at 6–8 months, 3–4 at 9–23 months, plus 1–2 snacks.
- **Cup size**: ½ cup at 6–12 months, ¾ cup at 12–24 months, 1 cup from 2 years.
- **"Mastitis: keep milk moving."**

- **Storage "4–4–6"**: 4 hours room, 4 days fridge, 6 months freezer.

OSCE tips

- The classic station is **"counsel a mother on breastfeeding"** or **"assess a breastfeed"**. Greet and sit at her level, **ask permission**, use **open questions**, **praise** what she does well (builds confidence), **observe before advising**, give **2–3 relevant suggestions** not a lecture, use the **4 positioning points and 4 attachment signs**, check understanding (teach-back), and arrange follow-up.
- For **"counsel on complementary feeding"**, ask the child's age, do a **24-hour recall**, give **age-specific frequency, amount, thickness and variety**, and mention **hygiene, responsive feeding and feeding during illness**.
- For **hand expression** or **formula preparation** stations, demonstrate and ask the mother to show you back.

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SELF-TEST

Flashcards

30 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	The WHO/UNICEF/Uganda MoH breastfeeding recommendations in one line	Initiate within **1 hour**, **exclusive** for **6 months**, complementary foods from **6 months**, continue breastfeeding to **2 years or beyond**.
2	What is allowed in exclusive breastfeeding?	Breast milk (including expressed) plus **ORS, drops and syrups** of vitamins, minerals or medicines. Nothing else: no water, glucose, herbs or formula.
3	4 signs of good attachment	**More areola above** than below the mouth; **mouth wide open**; **lower lip turned out**; **chin touching the breast**.
4	4 points of good positioning	Baby's **head and body in line**; held **close**; **facing the breast** (nose to nipple); **whole body supported**.
5	Signs of effective suckling	**Slow, deep sucks with pauses**, **audible swallowing**, cheeks rounded (not drawn in), baby releases the breast satisfied.
6	Signs a young breastfed baby is getting enough milk	**8–12 feeds/24 h**, **≥ 6 wet nappies/day** after day 5, birth weight regained by **10–14 days**, steady weight gain on the card.
7	Prolactin vs oxytocin in lactation	**Prolactin**: milk **production**, triggered by suckling, higher at night. **Oxytocin**: **let-down** reflex; inhibited by stress, pain and doubt.
8	Why is colostrum important?	Rich in **secretory IgA, lactoferrin, white cells and vitamin A**; laxative (clears meconium, less jaundice); the baby's 'first immunisation'. Never discard it.
9	Foremilk vs hindmilk	**Foremilk**: watery, quenches thirst. **Hindmilk**: **fat-rich**, energy and satiety. Let the baby finish one breast before switching.
10	Why start complementary foods at 6 months (not earlier, not later)?	Breast milk no longer meets **energy, iron, zinc and vitamin A** needs; infant developmentally ready. Earlier = infection and less breast milk; later = **faltering and iron deficiency**.
11	Energy needed from complementary foods (breastfed child)	About **200 kcal/day** at 6–8 months, **300** at 9–11 months, **550** at 12–23 months.
12	UCG/IMCI complementary feeding at 6–9 months	Thick porridge/mashed foods, **2–3 meals + 1–2 snacks**; start **2–3 tablespoons**, increase to **½ cup** (250 mL cup).
13	UCG/IMCI complementary feeding at 9–12 and 12–24 months	9–12 mo: **½ cup, 3–4 meals + 1–2 snacks**. 12–24 mo: **¾ cup, 3–4 meals + 1–2 snacks**. ≥ 2 years: **at least 1 full cup**.
14	WHO 2023 complementary feeding: dietary diversity	**Animal-source foods (meat, fish, eggs) daily**; **fruits and vegetables daily**; **pulses, nuts and seeds frequently**; minimise starchy staples.
15	WHO 2023: foods and drinks to avoid (6–23 months)	Foods high in **sugar, salt or trans fats**; **sugar-sweetened beverages**; **non-sugar sweeteners**; **limit 100% fruit juice**.
16	WHO 2023: milks for non-breastfed children	**6–11 months**: infant formula or full-fat animal milk. **12–23 months**: animal milk. **Follow-up formulas not recommended**; no sweetened milks.
17	Minimum dietary diversity: the 8 food groups	Breast milk; grains/roots/tubers/plantains; pulses/nuts/seeds; dairy; flesh foods; eggs; vitamin A-rich fruits and vegetables; other fruits and vegetables. **MDD = ≥ 5 of 8.**
18	Minimum meal frequency	**2** for breastfed 6–8 months; **3** for breastfed 9–23 months; **4** for non-breastfed 6–23 months (including ≥ 2 milk feeds).

#	Question	Answer
19	Uganda IYCF statistics (UDHS 2016 via MIYCAN 2021)	Early initiation about **66%**, EBF about **66%**, continued BF at 2 years about **43%**, minimum acceptable diet only about **15%**.
20	HIV and breastfeeding: Uganda MIYCAN 2021	EBF 6 months on maternal ART; then complementary foods + BF. **HIV-negative exposed infant: stop at 12 months** with ≥ 500 mL/day other milk. **HIV-positive infant: continue to 24 months.** Stop gradually over 1–2 weeks.
21	WHO 2016 duration of breastfeeding for mothers living with HIV	**At least 12 months**, and may continue **up to 24 months or longer** (like the general population) with ART adherence support.
22	Why is mixed feeding dangerous under 6 months (especially with HIV)?	Other foods and fluids **inflame the infant gut mucosa** (and add contamination), letting HIV and pathogens cross; also reduce milk supply.
23	BFHI Step 4 and Step 6	**Step 4**: immediate, uninterrupted **skin-to-skin** and early initiation. **Step 6**: **no food or fluids other than breast milk** unless medically indicated.
24	BFHI Steps 7–10	**7** rooming-in 24 h; **8** responsive feeding (recognise cues); **9** counsel on **bottles, teats and pacifiers**; **10** coordinated discharge and ongoing support.
25	Management of mastitis	**Keep breastfeeding/expressing** from the affected side, rest, fluids, analgesia; if bacterial: **flucloxacillin/cloxacillin 500 mg orally 6-hourly for 10–14 days**. Abscess → drainage.
26	Engorgement management	**Frequent feeding** with good attachment, **hand-express** to soften the areola, **warm compress before** and **cold after** feeds, paracetamol.
27	Storage of expressed breast milk (CDC)	**Room temperature up to 4 h**, **fridge up to 4 days**, **freezer about 6 months** (up to 12 acceptable); use thawed milk within 24 h; never refreeze.
28	True contraindications to breastfeeding	Infant **classic galactosaemia**; maternal **Ebola**; certain **drugs** (cytotoxics, radioactive iodine); herpes lesion on the breast (that side). **Not** HIV on ART, hepatitis B, TB on treatment, malaria.
29	Safe formula preparation (when truly needed)	Boiled water at **≥ 70 °C**, exact level scoops (usually **1 per 30 mL**), no extra powder or sugar, **cup** feeding, discard after **1 hour**.
30	Feeding during and after illness	**Breastfeed more often**, extra fluids, soft favourite foods in small frequent amounts; never withhold food; **one extra meal a day for 2 weeks** after illness.

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. A 3-week-old girl (birth weight 3.1 kg, now 3.4 kg) is breastfed 10 times a day. Her mother complains of severe nipple pain since day 2 and a cracked left nipple. On observation the baby's mouth is not wide open, the lower lip is turned in and more areola is visible below the mouth than above. What is the MOST appropriate advice?

A Stop breastfeeding on the left side for 2 weeks

B Apply a nipple cream and continue the same technique

C Start formula top-ups after each feed

D Help her improve attachment (wide mouth, lower lip out, chin to breast, more areola above), start on the less sore side, and rub hindmilk on the nipple

E Give the baby nystatin drops and the mother miconazole cream

2. A mother of a 2-month-old in Gulu says she gives water 3 times a day 'because it is very hot and the baby is thirsty'. The baby is exclusively breastfed otherwise and growing along the median. What is the BEST advice?

A Continue the water but boil it first

B Stop the water; breast milk contains all the water a baby under 6 months needs, so breastfeed more often in hot weather

C Give ORS instead of water

D Replace water with diluted fruit juice

E Give glucose water only at night

3. A 10-month-old boy (WLZ −1.6, MUAC 13.1 cm) is breastfed and eats thin maize porridge twice a day. What is the MOST appropriate counselling according to UCG 2023/IMCI and WHO 2023?

A Continue the same porridge but add sugar for energy

B Stop breastfeeding so that he eats more porridge

C Give 3–4 meals of ½ cup plus 1–2 snacks, make porridge thick and enrich it with groundnut paste, milk or oil, and add an animal-source food and fruits or vegetables daily

D Start follow-up formula 500 mL daily

E Give the porridge by bottle so he finishes it

4. A 26-year-old woman living with HIV, on TLD with a viral load < 50 copies/mL at 34 weeks, has just delivered at Mengo Hospital. She asks how to feed her baby. What is the MOST appropriate advice in Uganda?

A Exclusive breastfeeding for 6 months, then complementary foods with continued breastfeeding, while she continues ART and viral load monitoring

B Exclusive formula feeding, because breastfeeding transmits HIV

C Breastfeed for 3 months, then switch abruptly to cow's milk

D Breastfeed and give formula top-ups so the baby gets less breast milk

E Express and heat-treat all breast milk for 2 years

5. A 12-month-old HIV-exposed boy has a negative DNA PCR at 9 months. His mother is on ART with a suppressed viral load. According to Uganda's MIYCAN 2021 guidance, what is the MOST appropriate plan? Advanced

A Stop breastfeeding abruptly today and test in 6 weeks

B Continue breastfeeding to 24 months; no further testing needed

C Stop breastfeeding at 6 months next time

D Encourage gradual cessation of breastfeeding over 1–2 weeks, give at least 500 mL/day of other milk, and do a DNA PCR 6 weeks after complete cessation

E Start replacement formula and continue breastfeeding at night

6. On day 4 after delivery, a breastfeeding mother has fever 38.7 °C and a red, hot, tender wedge-shaped area on her right breast. The baby is feeding well. She was told by a relative to stop breastfeeding from that side. What is the BEST advice?

A Stop breastfeeding on the right side until the redness settles

B Stop breastfeeding completely and start formula

C Incision and drainage today

D Apply cold compresses only; antibiotics are never needed

E Continue breastfeeding or expressing from the right side, rest, analgesia, and flucloxacillin 500 mg 6-hourly for 10–14 days if bacterial or not better in 24 h

7. A 6-day-old exclusively breastfed baby is brought in lethargic and jaundiced. Birth weight 3.3 kg, now 2.85 kg. Few wet nappies. Na 156 mmol/L. What is the MOST likely underlying problem? Advanced

A Breast-milk jaundice

B Insufficient breast milk intake (breastfeeding failure) with hypernatraemic dehydration

C Galactosaemia

D Hypothyroidism

E Normal physiological weight loss

8. Which of the following practices on a maternity ward is CONSISTENT with the Baby-Friendly Hospital Initiative Ten Steps?

A Giving glucose water to all babies born by caesarean section until the mother is awake

B Keeping babies in the nursery at night so mothers can rest

C Distributing free formula samples from a company to mothers at discharge

D Teaching mothers to bottle-feed expressed milk

E Placing the baby skin-to-skin with the mother immediately after birth and helping her initiate breastfeeding within the first hour

9. A 15-month-old girl has stopped breastfeeding because her mother is pregnant. The family asks which milk to give. According to WHO 2023, which is MOST appropriate?

A Pasteurised full-fat animal (cow's) milk, about 2 cups a day, alongside a diverse family diet

B Follow-up ('growing-up') formula

C Sweetened condensed milk diluted with water

D Flavoured milk drinks to encourage intake

E No milk; porridge is enough

10. A 4-month-old's mother returns to work at a market in Kampala and asks how to keep breastfeeding. What is the BEST advice?

A Start formula during the day and breastfeed at night

B Introduce porridge now because the baby is 'big enough'

C Give cow's milk by bottle while she is away

D Express breast milk before leaving (stores about 4 hours at room temperature), have the caregiver give it by cup, and breastfeed often when together, including at night

E Stop breastfeeding to avoid confusion

11. An 11-month-old's 24-hour recall: breastfed 5 times; millet porridge made with milk (morning); matooke with beans (lunch); posho with dodo (amaranth leaves) (evening). No snacks. Which statement is CORRECT?

Advanced

A He meets both minimum dietary diversity (5 of 8 food groups) and minimum meal frequency, so he has a minimum acceptable diet

B He fails minimum dietary diversity because he ate no meat, fish or eggs

C He needs 4 meals a day to meet minimum meal frequency

D Breast milk does not count as one of the food groups

E He fails minimum meal frequency because no snacks were given

12. A mother brings her 5-month-old baby, who is exclusively breastfed and gaining weight along the median line, asking to start porridge because 'he cries a lot in the evening'. What is the MOST appropriate response?

A Start thin porridge once daily

B Start formula in the evening

C Reassure her that weight gain shows he is getting enough, explain evening cluster feeding, advise feeding on demand, and plan complementary foods at 6 months

D Give gripe water in the evening

E Refer to a paediatrician for colic

13. A 7-week-old baby has creamy white plaques on the tongue and cheeks that do not wipe off. His mother reports burning nipple pain that began last week after 6 pain-free weeks of breastfeeding. What is the BEST management?

A Stop breastfeeding until the plaques disappear

B Nystatin oral suspension for the baby and miconazole (or nystatin) cream to the mother's nipples, continuing breastfeeding

C Treat the baby only

D Flucloxacillin for the mother

E Gentian violet to the baby's mouth for 4 weeks and switch to bottle feeding

14. Which of the following is a TRUE contraindication to breastfeeding?

A Maternal HIV infection with suppressed viral load on ART

B Maternal hepatitis B surface antigen positivity (baby vaccinated at birth)

C Maternal pulmonary TB on treatment for 3 weeks

D Classic galactosaemia in the infant

E Maternal malaria treated with artemether-lumefantrine

15. A mother who has to use formula (her infant has galactosaemia ruled out; she is on chemotherapy) asks how to prepare feeds safely. Which instruction is CORRECT?

A Add an extra scoop so the baby grows faster

B Use unboiled tap water to save time

C Use a bottle with a large teat so the baby feeds quickly

D Keep prepared milk at room temperature for the next day

E Use water boiled and still at least 70 °C, measure level scoops exactly as on the tin, feed by cup, and discard leftovers after 1 hour

Answers & explanations

1. D · Help her improve attachment (wide mouth, lower lip out, chin to breast, more areola above), start on the less sore side, and rub hindmilk on the nipple

D is correct: pain from day 2 with fissures and poor attachment signs means the cause is poor attachment; correcting it is the treatment, plus starting on the less sore side and hindmilk on the nipple. A is wrong: stopping the side for 2 weeks risks engorgement, mastitis and falling supply (short-term expressing from that side is acceptable if pain is severe). B is wrong: creams do not fix the cause. C is wrong: her weight gain is adequate; formula undermines supply and exclusive breastfeeding. E is wrong: thrush typically causes burning pain after a period of comfortable feeding, with oral white patches; here the problem is attachment.

2. B · Stop the water; breast milk contains all the water a baby under 6 months needs, so breastfeed more often in hot weather

B is correct: breast milk (especially foremilk) supplies enough water under 6 months, even in hot climates; extra water displaces milk, adds infection risk and breaks exclusive breastfeeding. A is wrong: boiled water is still unnecessary and reduces breastfeeding. C is wrong: ORS is only for diarrhoea or dehydration. D is wrong: juice is not recommended and breaks EBF. E is wrong: glucose water reduces suckling and supply.

3. C · Give 3–4 meals of ½ cup plus 1–2 snacks, make porridge thick and enrich it with groundnut paste, milk or oil, and add an animal-source food and fruits or vegetables daily

C is correct: at 9–12 months the UCG recommends ½ cup at each of 3–4 meals plus 1–2 snacks; thin porridge is energy-poor, so enrich it, and WHO 2023 recommends animal-source foods and fruits/vegetables daily. A is wrong: WHO 2023 advises against foods high in sugar. B is wrong: breastfeeding should continue to 2 years or beyond. D is wrong: follow-up formulas are not recommended and are costly. E is wrong: bottles cause infection and are discouraged; use a cup and spoon with responsive feeding.

4. A · Exclusive breastfeeding for 6 months, then complementary foods with continued breastfeeding, while she continues ART and viral load monitoring

A is correct: breastfeeding is recommended for all women irrespective of HIV status (MIYCAN 2021); with suppressed viral load on ART, transmission risk is very low, while not breastfeeding raises infant mortality. B is wrong: formula is not the default and carries high risks in Uganda unless strictly safe and sustainable. C is wrong: early, abrupt cessation increases risk (and is outdated advice). D is wrong: mixed feeding under 6 months increases HIV transmission and diarrhoea. E is wrong: heat-treating is not routine practice and is impractical.

5. D · Encourage gradual cessation of breastfeeding over 1–2 weeks, give at least 500 mL/day of other milk, and do a DNA PCR 6 weeks after complete cessation

D is correct: MIYCAN 2021 advises HIV-exposed, HIV-negative infants to stop breastfeeding at 12 months, gradually over 1–2 weeks, with ≥ 500 mL/day of alternative milk (cow, goat or soya), and EID requires a PCR 6 weeks after complete cessation (WHO allows continuation to 24 months with support). A is wrong: abrupt weaning causes breast engorgement, distress and higher viral shedding risk. B is wrong: continuing to 24 months is MIYCAN's advice for HIV-positive infants, and a final test after weaning is always required. C is wrong: the child is already 12 months. E is wrong: mixing breast and replacement feeding prolongs exposure without benefit.

6. E · Continue breastfeeding or expressing from the right side, rest, analgesia, and flucloxacillin 500 mg 6-hourly for 10–14 days if bacterial or not better in 24 h

E is correct: mastitis is treated by continued effective milk removal plus rest, fluids and analgesia; antibiotics (flucloxacillin/cloxacillin 500 mg 6-hourly for 10–14 days) if fever, fissures, or no improvement within 24 h. A is wrong: milk stasis worsens mastitis and can lead to abscess. B is wrong: stopping is unnecessary and harmful. C is wrong: drainage is for an abscess (fluctuant mass), not mastitis. D is wrong: a febrile mother with probable bacterial mastitis often needs antibiotics.

7. B · Insufficient breast milk intake (breastfeeding failure) with hypernatraemic dehydration

B is correct: a 13.6% weight loss, few wet nappies, hypernatraemia and jaundice in the first week point to inadequate intake (poor attachment/infrequent feeds). Rehydrate slowly, support lactation, supplement with EBM or formula by cup, check bilirubin and exclude sepsis. A is wrong: breast-milk jaundice occurs in a well, thriving baby beyond 2 weeks. C is wrong: galactosaemia causes vomiting, jaundice, hepatomegaly and E. coli sepsis, but not this pattern of dehydration with hypernatraemia from low intake. D is wrong: hypothyroidism causes prolonged jaundice and lethargy without acute weight loss. E is wrong: normal loss is up to about 7–10%, regained by 10–14 days.

8. E · Placing the baby skin-to-skin with the mother immediately after birth and helping her initiate breastfeeding within the first hour

E is correct: BFHI Step 4 requires immediate, uninterrupted skin-to-skin contact and early initiation. A is wrong: Step 6 forbids non-breast-milk feeds unless medically indicated; mothers after CS can do skin-to-skin in recovery. B is wrong: Step 7 requires rooming-in 24 hours a day. C is wrong: free samples violate the International Code (Step 1a) and Uganda's marketing regulations. D is wrong: Step 9 advises counselling about the risks of bottles and teats; use a cup.

9. A · Pasteurised full-fat animal (cow's) milk, about 2 cups a day, alongside a diverse family diet

A is correct: for non-breastfed children 12–23 months, WHO 2023 recommends animal milk (pasteurised, fermented or reconstituted evaporated milk) and states follow-up formulas are not recommended; non-breastfed children need about 2 or more milk feeds a day. B is wrong: follow-up formulas are not recommended. C is wrong: condensed milk is very high in sugar and not appropriate. D is wrong: flavoured or sweetened milks should not be used. E is wrong: milk is an important source of protein, calcium and energy, especially when other animal-source foods are scarce.

10. D · Express breast milk before leaving (stores about 4 hours at room temperature), have the caregiver give it by cup, and breastfeed often when together, including at night

D is correct: expressing and cup-feeding preserves exclusive breastfeeding to 6 months; expressed milk keeps about 4 hours at room temperature (CDC), longer in a fridge, and frequent feeds when together maintain supply. A is wrong: formula makes it mixed feeding and reduces supply. B is wrong: complementary foods start at 6 months. C is wrong: cow's milk is unsuitable under 6 months (and bottles add infection risk). E is wrong: working mothers can and should continue breastfeeding.

11. A · He meets both minimum dietary diversity (5 of 8 food groups) and minimum meal frequency, so he has a minimum acceptable diet

A is correct: his food groups are breast milk, grains/roots/plantains (millet, matooke, posho), pulses (beans), dairy (milk) and vitamin A-rich vegetables (dark green dodo leaves) = 5 of 8, meeting MDD; 3 meals meets MMF for a breastfed 9–23-month-old; together that is a minimum acceptable diet. WHO 2023 would still advise adding an animal-source food (egg, fish, meat) daily. B is wrong: MDD counts any 5 of the 8 groups; flesh foods and eggs are recommended but not mandatory for the indicator. C is wrong: 4 feeds is the MMF for non-breastfed children. D is wrong: since the 2021 WHO/UNICEF revision, breast milk is one of the 8 groups. E is wrong: snacks are not required to meet MMF.

12. C · Reassure her that weight gain shows he is getting enough, explain evening cluster feeding, advise feeding on demand, and plan complementary foods at 6 months

C is correct: steady growth shows adequate milk; evening fussiness and cluster feeding are normal; complementary foods start at 6 months (180 days) per WHO 2023. A is wrong: early porridge displaces breast milk and adds infection risk. B is wrong: formula breaks exclusive breastfeeding. D is wrong: gripe water is unnecessary and breaks EBF. E is wrong: a thriving baby with normal crying does not need referral.

13. B · Nystatin oral suspension for the baby and miconazole (or nystatin) cream to the mother's nipples, continuing breastfeeding

B is correct: oral thrush with new burning nipple pain after comfortable feeding = candida in both; treat mother and baby together and continue breastfeeding. A is wrong: there is no need to stop. C is wrong: treating only one partner causes re-infection. D is wrong: flucloxacillin treats bacterial mastitis, not candida. E is wrong: prolonged gentian violet can ulcerate the mouth, and bottles worsen the problem.

14. D · Classic galactosaemia in the infant

D is correct: infants with classic galactosaemia cannot metabolise galactose from lactose and need a galactose-free formula. A is wrong: in Uganda, mothers on ART are encouraged to breastfeed. B is wrong: hepatitis B is not a contraindication when the baby receives the birth-dose vaccine (and HBIG where available). C is wrong: mothers on TB treatment can breastfeed; the baby receives TB preventive therapy. E is wrong: malaria and artemether-lumefantrine are compatible with breastfeeding.

15. E · Use water boiled and still at least 70 °C, measure level scoops exactly as on the tin, feed by cup, and discard leftovers after 1 hour

E is correct: water at ≥ 70 °C kills Cronobacter and other bacteria; exact measuring avoids hypernatraemia or malnutrition; cups are easier to clean; leftovers are discarded within 1 hour (UCG). A is wrong: extra powder causes hypernatraemic dehydration and obesity. B is wrong: unboiled water causes diarrhoea. C is wrong: bottles are hard to clean and fast flow risks choking and overfeeding. D is wrong: prepared formula grows bacteria quickly at room temperature.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: 'My milk is not enough': a 6-week-old at a HC IV

Nalwoga brings her 6-week-old son, Isaac, to the young child clinic at a HC IV in Mpigi for his 6-week vaccines. She says her milk 'is not enough' and she has started giving cow's milk by bottle twice a day. Birth weight 3.2 kg.

1. What will you ask?

> Danger signs (feeding poorly? fever? convulsions?); **when breastfeeding started**, pre-lacteal feeds; **how often** by day and **night** (she says 6 times, none at night because he sleeps); what else and from when (cow's milk from 4 weeks, bottle); **wet nappies** (4–5/day); **breast pain** (sore nipples); her **HIV status** (negative at ANC, retest due), her health and mood, work plans, who helps at home.

2. What will you examine?

> **Weigh and plot**: 3.9 kg (gain of 700 g in 6 weeks ≈ 17 g/day, below the expected 25–30 g/day; WAZ falling). **Observe a breastfeed**: body turned away, neck twisted, **chin off the breast, mouth not wide open, more areola below**, quick shallow sucks. Mouth: no thrush, no tongue-tie. Breasts: small fissure on the right nipple. Baby alert and well.

3. What is your assessment?

> **Feeding problem**: **poor positioning and attachment**, **too few feeds (6/day, none at night)**, and **mixed feeding with a bottle**, causing **slow weight gain** and reduced milk supply. No danger signs; not a true milk insufficiency.

4. How do you counsel her?

> Praise her for coming and for breastfeeding. Show **positioning** (in line, close, facing, supported) and **attachment** (wide mouth, lower lip out, chin to breast, more areola above); practise until comfortable. Feed **8–12 times in 24 h including at night**; let him finish one breast first. **Stop cow's milk and the bottle** gradually as supply rises (cow's milk is unsuitable under 12 months as a main drink, and mixed feeding raises diarrhoea risk). Skin-to-skin. Rub hindmilk on the sore nipple. Check understanding by teach-back.

5. What else do you do today, and when do you review?

> Give the **6-week vaccines** (OPV1, DPT-HepB-Hib1, PCV1, rotavirus 1). **Retest the mother for HIV** (postnatal retesting). Discuss **family planning** (LAM requires exclusive breastfeeding, which she is not yet doing; offer a progestogen-only method). **Review in 5 days** (IMCI feeding problem) to observe a feed and weigh; she should see **more wet nappies** within days.

6. At review she is exclusively breastfeeding 10 times a day, and he has gained 180 g in 5 days. What do you say?

> Excellent progress: **36 g/day**. Praise her and her family; continue exclusive breastfeeding to 6 months; teach **expressing and cup feeding** for when she returns to work; plan complementary feeding at 6 months; return for 10-week vaccines.

Take-away: Most 'insufficient milk' is poor attachment and infrequent feeding: watch a feed, fix positioning and attachment, feed often day and night, stop supplements, and review weight within a week.

Case 2: HIV-exposed infant: feeding through the first two years

Akello, 24, is living with HIV and on TLD. Her viral load at 36 weeks was < 50 copies/mL. She delivers a healthy 3.0 kg girl, Grace, at a regional referral hospital. She is worried that breastfeeding will infect her baby.

1. What will you tell her about breastfeeding and HIV?

> With her viral load suppressed on ART, the risk of passing HIV through breast milk is **very low**; not breastfeeding greatly raises Grace's risk of dying from **diarrhoea, pneumonia and malnutrition** in Uganda. Uganda recommends **breastfeeding for all women irrespective of HIV status** (MIYCAN 2021): **exclusive breastfeeding for 6 months**, keep taking ART every day, keep her breasts healthy, and attend viral load checks.

2. What does Grace need in the first weeks?

> **Infant prophylaxis**: low-risk (mother suppressed) → **NVP once daily for 6 weeks** (15 mg if ≥ 2.5 kg) started within 6 h of birth; **co-trimoxazole** from 6 weeks; **DNA PCR at 4–6 weeks**; routine vaccines. **Early initiation** of breastfeeding within 1 hour; skin-to-skin; no pre-lacteal feeds.

3. At 3 months Akello plans to give water and porridge because her mother-in-law insists. What do you say?

> **No mixed feeding before 6 months**: other foods and fluids inflame the baby's gut and **increase HIV transmission** and diarrhoea. Involve the mother-in-law in counselling; explain that breast milk provides all the water and food Grace needs until 6 months. Check Akello's **adherence and viral load** (every 3 months during breastfeeding).

4. At 6 months, what is the feeding plan?

> **Start complementary foods** (2–3 meals + 1–2 snacks, 2–3 tablespoons rising to ½ cup of thick, enriched porridge and mashed foods with an animal-source food, fruits and vegetables daily) **while continuing breastfeeding**. **Vitamin A 100,000 IU.** Continue maternal ART and VL monitoring. **DNA PCR at 9 months.**

5. Grace's 9-month PCR is negative. When and how should breastfeeding stop, and what testing follows?

> Uganda MIYCAN 2021: encourage stopping at **12 months** for HIV-negative exposed infants (WHO 2016 allows continuing up to 24 months with adherence support), **gradually over 1–2 weeks** (express and cup-feed, replace with other milk), then **at least 500 mL/day of cow's, goat's or soya milk** with a diverse diet. **DNA PCR 6 weeks after complete cessation**, then a **rapid antibody test at 18 months**.

6. What would change if Grace's PCR were positive?

> Start **ART immediately** (send a confirmatory sample at the same time), and **continue breastfeeding on demand to 24 months** (MIYCAN 2021) to protect her nutrition and health; intensive adherence support for both mother and baby; review Akello's viral load and adherence.

Take-away: In Uganda, an HIV-positive mother on suppressive ART breastfeeds exclusively for 6 months and continues with complementary foods, avoids mixed feeding, and stops gradually (about 12 months if the baby is negative) with a PCR 6 weeks after weaning.

Case 3: Growth faltering at 9 months: a complementary feeding consultation

Moses, 9 months old, is brought to the outpatient clinic at Mengo Hospital with a cough for 3 days. The nurse notes his weight (7.3 kg) has hardly changed since 6 months (7.0 kg). He is breastfed. His grandmother

looks after him while his mother sells vegetables.

1. After treating the cough (no pneumonia), how do you assess his nutrition?

> **Length 70 cm**: WLZ about −1.3, LAZ about −0.9, WAZ about −1.3; **MUAC 13.0 cm**; no oedema. The card shows **flattening since 6 months** = growth faltering. No danger signs. Check **Hb** (HemoCue 9.8 g/dL: mild anaemia).

2. What will you ask the grandmother?

> **24-hour recall**: breastfed 3 times (morning, evening, night); **thin maize porridge with sugar** by bottle twice; **mashed matooke with soup** once. Shares a plate with a 3-year-old cousin. **Food groups**: breast milk, grains/plantains = 2 of 8. **Meals**: 3. Also ask about **diarrhoea and malaria** episodes, **vitamin A and deworming** (not received), immunisation (MR due at 9 months).

3. Interpret the recall.

> **MMF met (3 meals for a breastfed 9-month-old)** but **MDD not met (2 of 8)**; low energy density (thin porridge), **no animal-source food**, no fruits or vegetables, **sugary** porridge, **bottle use**, **no own plate**, and breastfeeding has fallen. So **no minimum acceptable diet** → faltering and probable **iron deficiency**.

4. Give your counselling, specific to his age and family.

> - **Breastfeed** whenever with mother, including more at night; the mother can **express** milk for the day
> - **3–4 meals of ½ cup + 1–2 snacks**; make porridge **thick**, enrich with **groundnut paste, milk or oil** (not sugar)
> - **Every day**: an **animal-source food** (egg, silverfish powder, liver, milk) and **fruits/vegetables** (dodo, pawpaw, mango, orange sweet potato, avocado); **beans** mashed often
> - **His own bowl**; sit with him and encourage (**responsive feeding**)
> - **Cup and spoon**, no bottle; **hand-washing** with soap
> - Extra food and fluid during and after the cough

5. What preventive care does he need today?

> **Measles–rubella vaccine** (9 months) and any other catch-up; **vitamin A 100,000 IU**; deworming is from **12 months** (albendazole 200 mg); treat or investigate anaemia (iron-rich foods; iron supplement per programme; malaria test); **review weight in 2–4 weeks** and enrol in monthly growth monitoring.

6. How will you check that the grandmother has understood?

> **Teach-back**: ask her to say how many meals, how much, how thick, and which extra foods she will add tomorrow, using foods she has at home; agree on **one or two changes** she is confident about; give the mother the same messages (by phone or at the next visit).

Take-away: Growth faltering after 6 months is usually a complementary-feeding problem: do a 24-hour recall, count meals and food groups, and give age-specific, practical, local advice to the person who actually feeds the child.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.